

RESEARCH INTERESTS

I study the **theoretical foundations of large language models**: the algorithms they learn and how to steer them.

- **Algorithmic perspectives on LLMs**: whether Transformers learn algorithms from data [C4, C14], how pretrained LLMs do arithmetic [C3, C10, C18], and what sets Transformers apart from other architectures [C7].
- **Interpretability and alignment**: decision making under uncertainty [C6, C15], steering vectors [C12, C13], and mechanistic interpretability with SAEs and transcoders [C16, T1].
- **Foundation models beyond text**: multimodal reward models [C8, C5], modality sensitivity [C2], data for visual reasoning [C11], and tabular foundation models [T2, P4].

EDUCATION

University of Southern California

2022 – Present

Doctor of Philosophy (Ph.D.) in Computer Science

Advisors: Vatsal Sharan & Robin Jia

University of Chicago

2020 – 2022

Master of Science (S.M.) in Statistics

University of Chicago

2016 – 2020

Bachelor of Science (S.B.) in Mathematics with Honors

Bachelor of Science (S.B.) in Computer Science with Honors

Bachelor of Arts (A.B.) in Statistics

EXPERIENCE

Google Research

May 2026 – Present

Student Researcher

- Hosts: Weihao Kong, Abhimanyu Das

Simons Institute at UC Berkeley

Fall 2024

Visiting Graduate Student

- Program: Modern Paradigms in Generalization

Meta Llama Team

May – Dec., 2024

AI Research Scientist Intern

- Host: Lawrence Chen; Peers: Tong Xiao, Rui Wang, and Pengchuan Zhang

Google

May – Aug., 2023

Software Engineering Intern

- Team: Bard (now Gemini) Multilingual

PUBLICATIONS

featured * equal contribution † mentored student

[C19] Value-Aware Stochastic KV Cache Eviction for Reasoning Models

Ting-Yun Chang*, Harvey Yiyun Fu*, Deqing Fu*, Chenghao Yang, Jesse Thomason, Robin Jia

Neural Information Processing Systems (NeurIPS), 2026

[C18] Convergent Evolution: How Different Language Models Learn Similar Number Representations

Deqing Fu, Tianyi Zhou, Mikhail Belkin, Vatsal Sharan, Robin Jia

Conference on Language Modeling (COLM), 2026

[C17] Token-Level Off-Policy Learning for Faithful Generation Under Distribution Shift

Cynthia Huang*†, Gustavo Lucas Carvalho*†, Deqing Fu, Robin Jia

Conference on Language Modeling (COLM), 2026

[C16] Resa: Transparent Reasoning Models via SAEs

Shangshang Wang, Julian Asilis, Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Deqing Fu, Willie Neiswanger

Conference on Language Modeling (COLM), 2026

- [C15] **Are LLM Decisions Faithful to Verbal Confidence?**
Jiawei Wang, Yanfei Zhou, Siddartha Devic, Deqing Fu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026
- [C14] **Transformers Provably Learn Algorithmic Solutions for Graph Connectivity, But Only with the Right Data**
Qilin Ye[†], Deqing Fu^{*}, Robin Jia, Vatsal Sharan
International Conference on Machine Learning (ICML), 2026
- [C13] **EPSVec: Efficient and Private Synthetic Text Generation via Dataset Vectors**
Amin Banayeeanzade, Qingchuan Yang, Deqing Fu, Spencer Hong, Erin Babinsky, Alf Samuel, Anoop Kumar, Robin Jia, Sai Praneeth Karimireddy
International Conference on Machine Learning (ICML), 2026
- [C12] **Textual Steering Vectors Can Improve Visual Understanding in Multimodal Large Language Models**
Woody Haosheng Gan[†], Deqing Fu^{*}, Julian Asilis^{*}, Ollie Liu^{*}, Dani Yogatama, Vatsal Sharan, Robin Jia, Willie Neiswanger
Association for Computational Linguistics (ACL), 2026
- [C11] **Zebra-CoT: A Dataset for Interleaved Vision Language Reasoning**
Ang Li^{*}, Charles L. Wang^{*}, Deqing Fu^{*}, Kaiyu Yue^{*}, Zikui Cai^{*}, Wang Bill Zhu^{*}, Ollie Liu^{*}, Peng Guo^{*}, Willie Neiswanger, Furong Huang, Tom Goldstein, Micah Goldblum
International Conference on Learning Representations (ICLR), 2026
- [C10] **FoNE: Precise Single-Token Number Embeddings via Fourier Features**
Tianyi Zhou, Deqing Fu, Mahdi Soltanolkotabi, Robin Jia, Vatsal Sharan
International Conference on Learning Representations (ICLR), 2026
- [C9] **VisualLens: Personalization through Task-Agnostic Visual History**
Wang Bill Zhu, Deqing Fu, Kai Sun, Yi Lu, Zhaojiang Lin, Seungwhan Moon, Kanika Narang, Mustafa Canim, Yue Liu, Anuj Kumar, Xin Luna Dong
Neural Information Processing Systems (NeurIPS), 2025
- [C8] **TLDR: Token-Level Detective Reward Model for Large Vision Language Models**
Deqing Fu, Tong Xiao, Rui Wang, Wang Zhu, Pengchuan Zhang, Guan Pang, Robin Jia, Lawrence Chen
International Conference on Learning Representations (ICLR), 2025
- [C7] **Transformers Learn Low Sensitivity Functions: Investigations and Implications**
Bhavya Vasudeva^{*}, Deqing Fu^{*}, Tianyi Zhou, Elliot Kau[†], You-Qi Huang[†], Vatsal Sharan
International Conference on Learning Representations (ICLR), 2025
- [C6] **DeLLMa: Decision Making Under Uncertainty with Large Language Models**
Ollie Liu^{*}, Deqing Fu^{*}, Dani Yogatama, Willie Neiswanger
International Conference on Learning Representations (ICLR), 2025
ICLR 2025 Spotlight (Top 5.1%)
- [C5] **DreamSync: Aligning Text-to-Image Generation with Image Understanding Feedback**
Jiao Sun^{*}, Deqing Fu^{*}, Yushi Hu^{*}, Su Wang, Royi Rassin, Da-Cheng Juan, Dana Alon, Charles Herrmann, Sjoerd van Steenkiste, Ranjay Krishna, Cyrus Rashtchian
Nations of the Americas Chapter of the ACL (NAACL), 2025
- [C4] **Transformers Learn to Achieve Second-Order Convergence Rates for In-Context Linear Regression**
Deqing Fu, Tian-Qi Chen, Robin Jia, Vatsal Sharan
Neural Information Processing Systems (NeurIPS), 2024
SoCalNLP Symposium 2023 Best Paper Award
- [C3] **Pre-trained Large Language Models Use Fourier Features to Compute Addition**
Tianyi Zhou, Deqing Fu, Vatsal Sharan, Robin Jia
Neural Information Processing Systems (NeurIPS), 2024
- [C2] **IsoBench: Benchmarking Foundation Models on Isomorphic Representations**
Deqing Fu^{*}, Ruohao Guo^{*}, Ghazal Khalighinejad^{*}, Ollie Liu^{*}, Bhuwan Dhingra, Dani Yogatama, Robin Jia, Willie Neiswanger
Conference on Language Modeling (COLM), 2024

[C1]

SCENE: Self-Labelled Counterfactuals for Extrapolating to Negative Examples

Deqing Fu, Ameya Godbole, Robin Jia

Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023

PRE-PRINTS, TECH REPORTS, AND BLOG POSTS

featured

* equal contribution

† mentored student

[P4]

TabFM-Auto: Self-Evolving Pipelines for Tabular Foundation Models

Deqing Fu, Huangyuan Su, Rajat Sen, Taman Narayan, Sujay Sanghavi, Abhimanyu Das, Weihao Kong

[P3]

Rewards Beyond Test Passing: Towards Generalizable Fidelity in LLM Debugging

Yunkai Zhan, Miaosen Chai, Shangshang Wang, Jike Zhong, Yuqing Yang, Deqing Fu, Serina Chang, Wang Bill Zhu

[P2]

Learning to GUIDE the Group: Self-Learning LLM Facilitation Across Multi-Party Tasks

Ziyi Liu, Tzu-Sheng Kuo, Jiale Liu, Deqing Fu, Nithum Thain, Crystal Qian

[P1]

Unsupervised Tool Evolution for Coding Agents in Data-Lake Analytics

Yuqing Yang, Yurong Liu, Deqing Fu, Manav Garg, Sravan Babu Bodapati, Fatma Ozcan, Alon Halevy, Jiani Zhang

[T2]

TabFM: A Zero-shot Foundation Model for Tabular Data

Weihao Kong, Erez Louidor Ilan, Shuxin Nie, Taman Narayan, Rajat Sen, Yichen Zhou, Deqing Fu, Samet Oymak, Abhimanyu Das

Technical Report

[T1]

Qwen3 Cross-layer Transcoders

Jakob Hansen, Deqing Fu, Shangshang Wang, John Carlsson

Technical Report

[B1]

Seeing Is Not Reasoning: How VLMs and Their Benchmarks Lean on Text

Harvey Yiyun Fu*, Chenghao Yang*, Ting-Yun Chang, Zhaowei Wang, Zhaofeng Wu, Jiawei Zhou, Deqing Fu

Blog Post

AWARDS

ACL Outstanding Reviewer Award	ACL, 2026
NeurIPS Top Reviewer	NeurIPS, 2025
Viterbi Best Research Assistant Award (<i>one per department</i>)	USC, 2025
USC Capital One Center (CREDIF) PhD Fellow	USC, 2025
SoCalNLP Symposium 2023 Best Paper Award	SoCalNLP, 2023
Provost’s Fellowship	USC, 2022
Susanne H. Rudolph Scholarship	UChicago, 2020–2021
Liew Family College Research Fellows Fund	UChicago, 2020
Dean’s List	UChicago, 2016–2020

INVITED TALKS

Emergent Algorithms in Foundation Models: Theory, Discovery, and Design	
Columbia NLP Seminar	Nov 5, 2026 (scheduled)
Cornell Tech ML Seminar	Nov 3, 2026 (scheduled)
Harvard ML Foundations Seminar	Oct 28, 2026 (scheduled)
ISI Natural Language Seminar	Sep 22, 2026
USC NLP Lunch	Sep 10, 2026
Theoretical Foundations of LLMs: Emergent Algorithms and Model Design	
Google Research Seminar	Aug 6, 2026
Algorithmic Perspectives on Understanding Transformers	
FlaNN Talk Seminar	Aug 25, 2025
UChicago/TTIC NLP Seminar	May 08, 2025
Duke NLP Seminar	Feb 12, 2025
Closing the Modality Gap: Benchmarking and Improving Visual Understanding in Multimodal LLMs	
Stanford NLP Seminar	May 22, 2025

Simplicity Bias of Transformers to Learn Low Sensitivity Functions

USC NLP Lunch

Feb 22, 2024

Aligning Text-to-Image Generation with Image Understanding Feedback

USC NLP Lunch

Oct 19, 2023

Transformers Learn to Achieve Second-Order Convergence for In-Context Linear Regression

USC NLP Lunch

Mar 29, 2023

SERVICES

Reviewer for conferences in

- **Machine Learning:** ICLR (2025–2027), NeurIPS (2024–2026), ICML (2025)
- **Natural Language Processing:** COLM (2024–2026), EMNLP (2023), ACL Rolling Review (2024–2026)
- **Computer Vision:** CVPR (2023–2025), ECCV 2022, ICCV 2025

Organizer of USC ML/NLP Joint Reading Group

Jan 2025 – May 2025

Organizer of USC AI Safety Club

Jan 2024 – Jan 2025

STUDENTS SUPERVISED

Undergraduate Students

Zitong Cynthia Huang (USC UG)

Sep 2025 – May 2026

Now: Ph.D. Student in Electronic and Computer Engineering at University of Southern California

Qilin Ye (USC UG)

Sep 2023 – Feb 2026

Now: Ph.D. Student in Computer Science at Stanford University

Woody Haosheng Gan (USC UG)

Sep 2023 – May 2026

Now: Ph.D. Student in Computer Science at Georgia Institute of Technology

You-Qi Huang (USC UG)

Sep 2023 – Mar 2024

Master's Students

Sophia Mirrashidi (USC MS)

Jan 2026 – Present

Gustavo Lucas Carvalho (USC MS)

Jan 2025 – May 2026

Elliot Kau (USC MS)

Sep 2023 – Mar 2024

Now: Software Engineer at Google

Tian-Qi Chen (USC MS)

Mar 2023 – Sep 2023

TEACHING EXPERIENCE

Teaching Assistant, CSCI 467: Introduction to Machine Learning

Spring 2025

University of Southern California, Los Angeles, CA

Teaching Assistant, CSCI 566: Deep Learning and its Applications

Spring 2023

University of Southern California, Los Angeles, CA

Teaching Assistant, BUSN 41301: Statistical Insight

Fall 2021

Booth School of Business, University of Chicago, Chicago, IL

Course Assistant, CMSC 31230: Fundamentals of Deep Learning

Fall 2021

University of Chicago, Chicago, IL

Course Assistant, MATH 20300: Analysis in $\mathbb{R}^n$ – I

Spring 2018

University of Chicago, Chicago, IL

SKILLS

Languages: C/C++, Python, L^AT_EX, MATLAB, R, Java, CUDA

Frameworks: PyTorch, TensorFlow, Keras, JAX, HuggingFace